

Information Security

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What is Data?

Data is a collection of facts, observations, measurements, or information that can be processed or communicated.

- Examples:
- Text
- Numbers
- Images
- Audio
- Video
- Sensor readings
- Location information
- Passwords
- Financial records

Analog Data

Analog data is continuous information that can have infinitely many values within a range.

Examples:

- Human voice Temperature Pressure Light intensity Heartbeat Radio signal

For example, temperature can continuously change:

25.1°C → 25.11°C → 25.111°C → ...

Example: When you speak to your friend:

Voice → Sound waves → Microphone → Digital data

The original voice is a physical phenomenon, but the computer needs to represent it digitally.

Digital Data

Digital data represents information using discrete values, commonly 0 and 1.

- Examples:
- Computer files Emails Digital photographs Videos Databases Passwords
- Digital documents

At the fundamental level:

- **Digital Data → Bits → 0s and 1s**
- Example:
- 01001000 01101001
- A computer interprets these bits according to an encoding scheme.

Analog vs Digital Data

Analog Data	Digital Data
Continuous	Discrete
Infinite possible values	Limited discrete values
More susceptible to noise	Easier to detect/correct errors
Voice, temperature	Files, emails, databases
Traditional telephone signal	Computer/network data

Why Digital if Nature is Analog?

Real World → Analog Signal → Sensor → Digital Conversion → Digital Processing → Transmission/Storage → Digital-to-Analog → Real World

Nature is Analog

Most physical phenomena change **continuously**:

- Sound → continuously varying air pressure
- Temperature → continuously varying
- Light → continuously varying intensity
- Human voice → continuous waveform
- Heartbeat → continuous electrical signal
- Radio waves → continuous electromagnetic waves
- For example, your voice isn't naturally a sequence of 0s and 1s.

Your voice → analog sound wave

Computers are Digital

Computers are electronic machines designed primarily to work with **discrete states**, commonly represented as:
0 and 1

So when we want a computer to process something from the physical world:

- **Analog information must be converted into digital data.**

For example:

Human voice



Microphone



Analog electrical signal



ADC (Analog-to-Digital Converter)



Digital data (0s and 1s)



Computer processing

Why Digital is Better for Information Security

Digital information has major advantages:

Analog	Digital
Continuous	Discrete
More affected by noise	More resistant to noise
Difficult to reproduce exactly	Can be copied exactly
Processing is harder	Easy for computers to process
Security mechanisms are limited	Encryption can be applied
Storage can degrade	Digital copies can remain identical
Error detection is difficult	Error detection/correction is possible

But Digital Systems Still Interact With Analog Nature

Digital technology doesn't replace the analog world; it creates a digital representation of the analog world.

For example, when you take a photograph:

- **Real scene (analog)**



Camera sensor



ADC



Digital pixels



JPEG/PNG file



Storage / Internet



Display



Light reaching your eyes (analog)

So digital is essentially a **convenient representation of reality**.

Why digital?

- Nature is analog, but computers are digital; therefore, we convert analog information into digital form so that computers can efficiently process, store, transmit, and secure it.

Processing Chain:

**Analog → ADC → sampling → quantization → binary representation
→ networking → information security.**

Analog-to-Digital Conversion

How does a smartphone convert your voice into digital information?

Basic process:

Sound



Microphone



Analog Electrical Signal



Sampling



Quantization



Encoding



Digital Data

- **Sampling:** The continuous signal is measured at regular intervals.
- **Quantization:** Each sample is assigned a numerical value.
- **Encoding:** Those values are represented digitally as bits.

Why Digital Data is Important for Security

- Once information becomes digital, it can be:
- Stored Copied Processed Transmitted Encrypted Compressed Analyzed

But it can also be:

- Stolen Modified Deleted Intercepted Destroyed
- This is where **Information Security** becomes important.

From Digital Data to Networking

- **What is a Network?**
- A **network** is a collection of connected devices that communicate and share information.
- Examples:
 - Computer network
 - Wi-Fi network
 - Mobile network
 - Internet
 - IoT network
- **Simple example**
- **Laptop → Wi-Fi Router → Internet → Server**

Why Do We Need Networking?

- Networking allows us to:
- Share information
- Access websites
- Send emails
- Make video calls
- Share files
- Use cloud services
- Access databases
- Communicate with IoT devices

How Does Digital Data Travel?

- Suppose Ali sends a message to Ahmed.

Simplified flow:

Ali's phone



Digital data



Network interface



Wi-Fi/Router



Internet



Ahmed's router



Ahmed's phone

- But the complete message is generally broken into smaller units for transmission.

Packets

- A **packet** is a unit of data transmitted over a network.
- A simplified packet can contain:
 - **Header + Data/Payload + Other control information**
 - **Header may contain information such as:**
 - Source address
 - Destination address
 - Protocol information
 - Sequence/control information
 - **Important distinction**
- **Data → divided into packets → transmitted → reconstructed**

Frames, Packets and Data

- **Application Data**



Transport Segment



Network Packet



Data-Link Frame



Bits/signals

- This is associated with different layers of network communication.

Networking Protocols

A **protocol** is a set of rules used for communication.

Examples: HTTP HTTPS TCP UDP IP DNS DHCP SSH

When you visit a website:

- **Browser → DNS → IP address → TCP/QUIC → HTTPS → Web server**

Where Does Information Security Come In?

- **What happens if someone intercepts the communication?**

Possible consequences:

- Reading confidential information
- Modifying information
- Stealing credentials
- Redirecting communication
- Disrupting communication

Therefore:

- **Networking enables communication.**
- **Information Security protects that communication.**

Example — Voice Message

Suppose Ali records a confidential voice message.

Step 1 — Creation

Voice

Step 2 — Conversion

Analog → Digital

Step 3 — Storage

Digital audio file

Step 4 — Protection

Encryption

Step 5 — Transmission

Network packets

Step 6 — Reception

Ahmed receives the data.

Step 7 — Reconstruction

Packets → Data → Audio → Sound

From Data to Security

Real-world information



Analog Data



ADC



Digital Data



Computer



Network



Packets



Internet



Receiver

At every stage **SECURITY IS REQUIRED**

Risk is everywhere!

Data exists → Data is converted → Data is transmitted → Data can be attacked

At every stage, different attacks can occur:

- **Analog stage:** signal interception, interference
- **Digital stage:** data modification or corruption
- **Network stage:** packet sniffing, spoofing, Man-in-the-Middle
- **Storage stage:** unauthorized access, data theft
- **Transmission stage:** eavesdropping and interception

What is Information Security?

- **Information Security is the practice of protecting information and information systems from unauthorized access, use, disclosure, disruption, modification, or destruction.**
- The main goal is to protect:
- **Information + Systems + Communication**

Why is Information Security Important?

Consider:

- Online banking
- University databases
- Hospital records
- Government systems
- E-commerce
- Social media
- Cloud systems
- Industrial systems

Almost every modern organization depends on digital information that's why information security is compulsory not an option.